



Hacettepe University
Computer Engineering Department
BBM479/480 End of Project Report

Project Details

Field	Information
Title	Multi-Frame Low-Resolution License Plate Recognition
Supervisor	Nazlı İkizler Cinbis

Group Members

#	Full Name	Student ID
1	Zaina Abushaban	2210765053
2	Sila Keskin	2210765032
3	Anar Mammadov	2220765063
4	Süleyman Yolcu	2210765016

Final report prepared for BBM479/480 End of Project evaluation

Abstract of the Project

Automatic license plate recognition is an important computer vision task used in intelligent transportation, traffic safety, access control, and urban surveillance systems. In controlled scenes, modern recognition systems can perform well because the license plate is close to the camera and characters are sharp. However, real-world surveillance footage often contains very small plates, motion blur, compression artifacts, rain, shadows, and changing illumination. These effects make low-resolution license plate recognition difficult, especially when visually similar characters such as 0/O, 1/l, 5/S, and 8/B must be separated from only a few pixels.

This project addresses multi-frame low-resolution license plate recognition using the LRLPR-26 competition setting. Each vehicle track contains five low-resolution frames, while the goal is to predict the full plate text and provide a confidence score. Instead of treating each image independently, our solution uses the complementary information available across frames and across several OCR models. The developed pipeline includes plate/frame preprocessing, normalization, sharpening and contrast enhancement, multiple OCR backbones, and a track-level ensemble module. The final prediction is not produced by simple majority voting only; it combines model confidence, agreement between models, best-frame selection, and character-level evidence collected across the track.

Our best validation-selected ensemble was the HGRDT 220 leaf63 setting. It reached 98.35% exact-match recognition and 99.64% character accuracy on validation, and achieved 79.17% exact-match recognition and 94.53% character accuracy on the competition test set. The difference between validation and test performance shows that the hidden test set contains harder domain conditions than the validation split, but the high character accuracy indicates that most predictions are close to correct and that the remaining errors are concentrated in a small number of ambiguous characters or frames. The project demonstrates that multi-frame evidence and ensemble fusion can significantly improve robustness for low-resolution license plate recognition. Future work should focus on stronger domain generalization, temporal attention, synthetic degradation training, and improved calibration of the confidence score.

Introduction, Problem Definition & Literature Review

1.1 Introduction to the Field

Automatic license plate recognition (ALPR) is a specialized visual recognition problem that usually consists of plate detection, plate rectification or cropping, and character recognition. It is used in parking systems, toll collection, traffic monitoring, stolen vehicle search, border control, and smart-city applications. A conventional ALPR pipeline is successful when the camera is close to the vehicle, the plate is frontal, and the image resolution is sufficient for individual characters to be read clearly. In many surveillance scenarios, however, cameras are placed far from the road and vehicles move quickly. The plate occupies only a very small region of the frame, and the readable information may be distributed across several video frames rather than appearing clearly in a single image.

Low-resolution license plate recognition is therefore different from ordinary scene text recognition. A scene text model can often rely on large character shapes and contextual information from words. A license plate, by contrast, is a short sequence of letters and digits with limited linguistic context. A single wrong character makes the whole plate incorrect. The task is also sensitive to local visual details: a small stroke may be the only difference between B and 8 or between O and 0. This makes exact-match accuracy much harder to optimize than character-level accuracy.

1.2 Problem Definition

The problem studied in this project is multi-frame low-resolution license plate recognition. For each vehicle track, the input consists of five low-resolution license plate frames. In the training setting, high-resolution plate images may also be available as hidden or auxiliary references, while in the final test setting only the low-resolution frames are used. The expected output is the predicted plate text and a confidence score. The main challenge is to recover a reliable sequence from degraded evidence, where some frames may be blurred, partially occluded, affected by lighting, or simply too small to contain clear character boundaries.

The project hypothesis is that a multi-frame and multi-model approach can outperform a single-frame or single-backbone recognizer. The reason is that different frames preserve different details, and different OCR backbones make different mistakes. By collecting evidence at the track level, the system can select more reliable frames, reduce random errors, and use agreement between models as an additional confidence signal.

1.3 Literature Review

Earlier ALPR systems often used handcrafted image processing: thresholding, edge detection, connected components, morphological operations, and template matching. These methods were interpretable and efficient, but they were highly sensitive to illumination, camera angle, and plate style. Deep learning shifted the field toward learned detection and recognition. Object detectors such as the YOLO family made real-time plate localization practical, while convolutional neural networks learned robust features for character recognition.

For sequence recognition, CRNN-style models introduced a strong baseline by combining convolutional feature extraction with recurrent sequence modeling and Connectionist Temporal Classification (CTC) loss (Shi et al., 2016). This is useful when characters are not segmented manually. Attention-based scene text recognizers later improved flexibility by decoding character sequences directly, but they can still fail under severe blur or domain shift. More recent visual backbones such as ResNet (He et al., 2016) and ConvNeXt (Liu et al., 2022) provide stronger feature extraction, while transformer-based text recognition systems model longer-range dependencies. However, most generic text-recognition methods are designed for images where the text is visible at a reasonable scale. Their limitation in our setting is that low-resolution plates contain weak local evidence, and the model can become overconfident even when one or two characters are uncertain.

Super-resolution methods attempt to reconstruct a high-resolution image before recognition. This is attractive because the training data may include high-resolution plate versions. Nevertheless, super-resolution is not always aligned with OCR accuracy: a visually sharper plate is not necessarily the plate that gives the correct characters. In preliminary experiments, super-resolution-style processing helped image quality but did not by itself solve the recognition problem. For this reason, our final direction emphasized OCR robustness and track-level decision fusion rather than relying only on image reconstruction.

Ensemble learning is a natural solution for this task. If several recognizers are trained with different backbones or preprocessing, their errors are not perfectly correlated. A simple majority vote can improve stability, but it ignores confidence and character-level evidence. Our final approach extends majority voting by combining confidence fusion, model agreement, and best-frame selection. This directly addresses the core limitation of single-model recognition: a single network may read one frame incorrectly, while the track as a whole can still contain enough evidence for the correct plate.

1.4 Proposed Solution Summary

The proposed system is a validation-selected ensemble for multi-frame low-resolution plate recognition. The pipeline receives five low-resolution frames, applies preprocessing and normalization, runs several OCR backbones in parallel, and then aggregates the predictions at track level. The ensemble uses exact prediction agreement, confidence scores, and character-wise evidence to produce the final plate text and confidence. The goal is not only to maximize validation accuracy but also to produce a robust prediction for hidden test tracks with diverse lighting and weather conditions.

Methodology

2.1 Dataset

The project uses the LRLPR-26 multi-frame license plate recognition dataset. According to the project setting used in our experiments, the training set contains 20,000 vehicle tracks and the test set contains 3,000 vehicle tracks. Each track contains five low-resolution plate images. In the training data, five high-resolution plate images are also available as hidden or auxiliary references, while the final test input contains only the five low-resolution images. The dataset contains two scenario groups: Scenario A includes 10,000 controlled daylight tracks, and Scenario B includes 10,000 tracks under more diverse conditions such as rain, shadows, and difficult illumination. The target label is the license plate text; the final submission also includes confidence.



Figure 1. Representative examples of low-resolution license plate images from the training dataset. The samples illustrate typical degradation effects including severe downsampling, blur, low contrast, compression artifacts, and loss of fine character details. Under these conditions, visually similar characters become increasingly difficult to distinguish.



Figure 2. Corresponding high-resolution reference images available in the training setting. These images preserve substantially more character structure and visual information and were used as auxiliary references during exploratory experiments and model development.



Figure 3. Example samples from the hidden test setting. Compared with the training references, test images exhibit stronger degradation and contain only low-resolution observations, making robust recognition significantly more challenging.

The visual examples highlight the main challenge addressed in this project. While training data may contain high-resolution reference images, the final recognition task must operate only on low-resolution inputs under degraded conditions. Character boundaries are often incomplete, blurred, or compressed, and important visual evidence may be distributed across multiple frames rather than appearing clearly in a single image. Furthermore, the hidden test set contains substantially more difficult examples with stronger information loss and reduced visual quality. These observations motivated the investigation of super-resolution methods during early development and ultimately supported the adoption of a multi-frame OCR and ensemble-based recognition strategy in the final methodology.

Dataset component	Description
Input per track	5 low-resolution plate frames
Training reference	5 high-resolution plate images available in training setting
Training tracks	20,000
Test tracks	3,000
Scenarios	A: controlled daylight; B: diverse weather/lighting
Output	Plate text and confidence score

2.2 Data Preprocessing

The preprocessing stage aims to reduce nuisance variation before OCR. All plate frames are converted to a consistent format and size expected by each model. Normalization is applied to stabilize the input distribution. Image enhancement is used carefully because aggressive enhancement may create artificial strokes and cause character confusion. The applied operations include denoising, sharpening, contrast enhancement, and resizing. These steps are especially important for frames where the plate is dark, compressed, or motion blurred.

During training and validation, the same train/validation split is used for all models. This is important because ensemble selection must be fair: if each model uses a different validation set, the ensemble would be optimized on inconsistent evidence. The shared split allows us to compare backbones, test voting strategies, and choose ensemble hyperparameters using the same validation tracks.

The dataset was divided at the track level into training and validation subsets using a fixed random seed to ensure reproducibility across experiments. Approximately 90% of the tracks were used for training and the remaining 10% were reserved for validation. Splitting was performed before model training so that all OCR backbones and ensemble configurations were evaluated under identical conditions. Track-level separation prevented frames belonging to the same vehicle sequence from appearing in both subsets and reduced the risk of data leakage.

2.3 OCR Backbones

The recognition stage uses several OCR backbones rather than a single architecture. The final project poster summarizes the main parallel backbones as ResNet34, ConvNeXt Tiny, ResNet50, and ConvNeXt Small. These models are suitable because they provide different feature representations and different capacity levels. ResNet models are strong convolutional baselines with residual connections, while ConvNeXt models use modernized convolutional blocks that often perform well on visual recognition tasks. In earlier development, we also explored temporal CNN, transformer-based recognition, SVTR-style architectures, ReSTRAN-style models, and Mamba-based sequence modeling alternatives. The final system focuses on the models that were stable and useful inside the ensemble.

Each OCR model produces a candidate plate string and a confidence value for each frame or track. Because the same vehicle appears in five frames, the model can either aggregate features internally or produce frame-level predictions that are later fused. The important design choice is that no single frame is assumed to be perfect. Instead, the ensemble treats all frame predictions as evidence and gives more weight to predictions that are supported by multiple models or by higher confidence.

2.4 Track-Level Ensemble

The central contribution of the methodology is the track-level ensemble. A simple majority vote chooses the most frequent full plate string. This is useful when several models agree exactly, but it fails when the correct answer is split across predictions. For example, one model may predict 34ABC123, another 34ABC128, and another 34ABC12B. Full-string majority may be weak, but character-level evidence can show that most characters are stable and only the final character is uncertain.

The final ensemble therefore uses three complementary signals. First, majority voting measures exact agreement between complete predictions. Second, confidence fusion uses model confidence values to reduce the influence of uncertain outputs. Third, character-level evidence compares the candidate strings position by position and selects the most supported character when full-string agreement is insufficient. A best-frame selection signal is also used when one frame is clearly sharper or receives consistently higher confidence across models.

The final plate prediction can be described as selecting the candidate sequence y that maximizes an aggregate score $S(y)$:

$$S(y) = \alpha * \text{Agreement}(y) + \beta * \text{Confidence}(y) + \gamma * \text{CharacterEvidence}(y)$$

Here, $\text{Agreement}(y)$ measures how many models or frames support the sequence, $\text{Confidence}(y)$ summarizes the confidence of the supporting predictions, and $\text{CharacterEvidence}(y)$ measures how strongly each character position is supported across the track. The weights are selected on the validation set. The selected final setting in our experiments was HGRDT 220 leaf63, which gave the best validation-selected trade-off before producing the final test submission.

2.5 Evaluation Principles

The project uses two main evaluation metrics. Exact-match recognition accuracy counts a track as correct only if the entire predicted plate text is identical to the ground truth. This is the strictest and most important metric for practical ALPR, because one wrong character may identify a different vehicle. Character accuracy measures the percentage of individual characters predicted correctly. It is less strict but useful for diagnosis, because it shows whether the model is generally reading the plate structure correctly even when a small number of characters are wrong.

Validation results are used for model selection, hyperparameter tuning, and ensemble selection. Test results are used only for final evaluation. This separation avoids tuning the method directly on test labels. In addition to accuracy, we analyze the gap between validation and test performance, because a large gap indicates domain shift, overfitting to validation conditions, or hidden test scenarios that are more difficult than the local split.



Multi-Frame Low-Resolution License Plate Recognition

By Zaina Abushaban, Sila Keskin, Anar Mammadov, Süleyman Yolcu

Supervisor: Nazlı İkizler Cinbiş

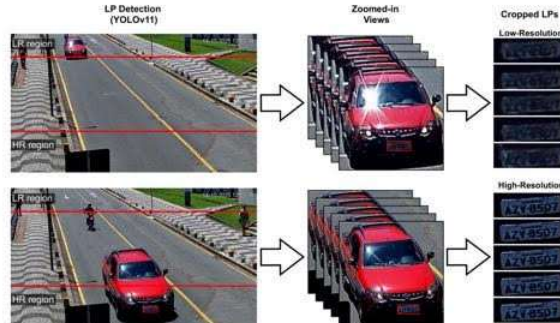
Project overview

What is our purpose?

- Address low-resolution license plate recognition under real-world surveillance conditions
- Build a multi-frame system that combines complementary OCR backbones and a track-level ensemble

The Challenge

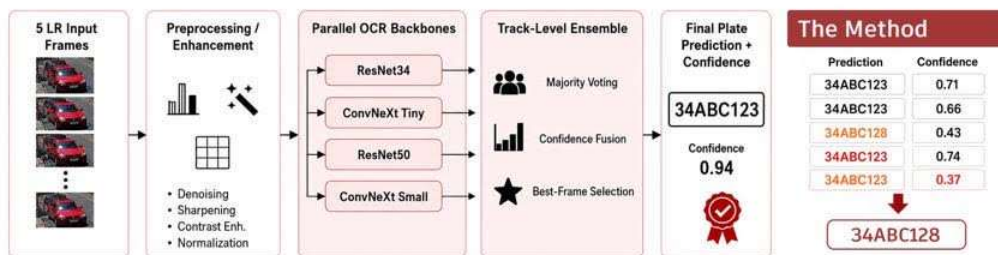
- Fast-moving cars cause motion blur.
- Plates appear tiny in real footage.
- Low resolution hides character details.
- Lighting, shadows, rain, and noise reduce clarity.
- Similar characters become hard to distinguish.



The Dataset



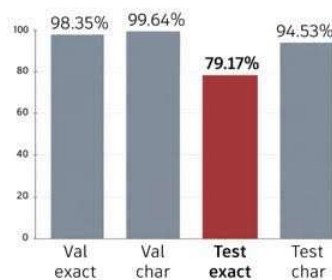
- LRLPR-26** benchmark
- Per training**
5 LR + 5 HR images
- 20,000** training tracks
- Test input**
5 LR images only
- 3,000** test tracks
- Output**
place text + confidence
- Scenario A**
10,000 controlled daylight tracks
- Scenario B**
10,000 diverse condition tracks



The final prediction is produced by a validation-selected ensemble instead of a simple majority vote. It combines predictions from multiple OCR backbones and uses confidence, model agreement, and character-level evidence to choose the most reliable plate output.

Ensemble experiments:

- HGRDT 09 leaf31
- HGRDT 160 leaf31
- HGRDT 220 leaf63 **SELECTED**
- ExtraTrees



Char - percentage of characters recognized correctly
Exact - percentage of plates recognized fully correctly

Results



Figure 4. Project poster summarizing the dataset, pipeline, ensemble method, and final performance.

Results & Discussion

3.1 Preliminary Super-Resolution Experiments on MDLP Dataset

Before establishing the final recognition pipeline, we conducted preliminary experiments to investigate whether super-resolution (SR) techniques could improve low-resolution license plate recognition performance. These experiments were performed on the MDLP dataset and evaluated several traditional interpolation-based approaches together with learning-based super-resolution models.

The motivation behind this stage was that recovering higher-resolution visual details could potentially improve OCR performance by making character boundaries more distinguishable. Since low-resolution license plates often contain blurred or compressed character regions, super-resolution appeared to be a promising preprocessing direction during the early stages of the project.

The evaluated methods included bilinear interpolation, Lanczos interpolation, bicubic interpolation, bicubic sharpening, bicubic with CLAHE enhancement, HAT, and ESRGAN. Performance was evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

Method	PSNR $\uparrow$	SSIM $\uparrow$
Bilinear	12.8117	0.2311
Lanczos	12.6720	0.2442
Bicubic	12.6671	0.2421
Bicubic + Sharpen	12.1323	0.2458
Bicubic + CLAHE	10.8250	0.2346
HAT	13.9481	0.2274
ESRGAN	11.5980	0.2730

Quantitative evaluation showed that reconstruction-oriented metrics and OCR-oriented performance were not always aligned. HAT achieved the highest PSNR value (13.95), while ESRGAN obtained the highest SSIM score (0.273), indicating stronger perceptual similarity. However, neither approach produced consistent improvements in downstream license plate recognition. This observation suggested that higher reconstruction quality does not necessarily preserve character-level information required for reliable OCR.



Figure 5. Example of Real-ESRGAN reconstruction on the MDLP dataset.

The original sequence is arranged from lowest quality (First) to highest quality (Last), with the ground-truth high-resolution image shown as reference (GT – Ground Truth). The lower row presents the corresponding super-resolved outputs. While super-resolution improved perceptual sharpness for higher-quality inputs, severely degraded frames still suffered from missing character information and reconstruction artifacts. In these cases, visually plausible details did not necessarily correspond to the original plate structure and could negatively affect character recognition reliability.

The results showed that super-resolution methods improved image appearance according to reconstruction metrics; however, these improvements did not consistently translate into better OCR performance. In particular, learning-based approaches occasionally generated visually plausible but non-original character details, introducing artificial structures and local distortions. Although such outputs sometimes appeared sharper to human observers, they could alter fine character boundaries and reduce recognition reliability.

Another observed limitation was information loss during reconstruction. Since the original low-resolution frames already contained limited character evidence, super-resolution models were unable to recover missing information reliably and occasionally amplified noise or generated ambiguous character strokes. This behavior increased the risk of confusing visually similar license plate characters.

Overall, these experiments suggested that optimizing image reconstruction quality alone was insufficient for improving recognition accuracy. As a result, super-resolution was not included in the final methodology, and subsequent development focused on improving OCR robustness through multi-frame evidence aggregation, ensemble learning, and track-level ensemble learning and confidence-based prediction fusion.

3.2 Validation and Test Performance

The final selected ensemble achieved strong validation performance and a competitive hidden-test result. On validation, the system reached 98.35% exact-match accuracy and 99.64% character accuracy. On the test set, the exact-match accuracy was 79.17%, while character accuracy was 94.53%. The high validation scores show that the recognition and ensemble pipeline learned the training-domain structure very well. The lower test exact-match score indicates that the hidden test split contains more difficult or shifted cases than the validation data.

Metric	Validation	Test
Exact-match plate accuracy	98.35%	79.17%
Character accuracy	99.64%	94.53%

The difference between exact-match and character accuracy is meaningful. A test character accuracy of 94.53% means that most characters are recognized correctly. However, exact-match accuracy drops because a plate is counted as wrong even if only one character is incorrect. This is expected for license plate recognition, where the sequence length is short but the exact full string is the final goal. The result suggests that the remaining failures are not completely wrong readings; many are likely near-misses caused by one ambiguous character, blur, or a difficult frame.

3.3 Ensemble Experiments

Several ensemble variants were tested. The project poster lists HGRDT 09 leaf31, HGRDT 160 leaf31, HGRDT 220 leaf63, and ExtraTrees as ensemble experiments. The selected setting was HGRDT 220 leaf63. This variant was chosen because it gave the best validation-selected performance and provided a good balance between agreement, confidence, and character-level correction. The selection process shows that the ensemble design mattered: simply collecting many predictions is not enough; the aggregation rule must be tuned to the task and validation distribution.

Experiment	Status / comment
HGRDT 09 leaf31	Tested ensemble candidate
HGRDT 160 leaf31	Tested ensemble candidate
HGRDT 220 leaf63	Selected final ensemble
ExtraTrees	Alternative ensemble candidate

3.3.1 Earlier Backbone and Ensemble Exploration

Model	Backbone	Validation Exact	Eval Exact
Confidence Ensemble	ResNet34 + ConvNeXt Tiny	–	75.53
ConvTranOCR	ConvNeXt Tiny	74.27	73.93
ResTranOCR	ResNet34	75.98	73.73
Confidence Ensemble	ResNet50 + ConvNeXt Small	–	74.90

Table shows summary of exploratory OCR and ensemble configurations evaluated during development. During development, several OCR architectures and confidence-based ensemble configurations were explored before selecting the final recognition pipeline. These experiments included different backbone combinations such as ResNet34, ResNet50, ConvNeXt Tiny, and ConvNeXt Small together with ResTranOCR, ConvTranOCR, and confidence-based ensemble approaches. The goal of this stage was to evaluate whether performance improvements could be achieved through model diversity and prediction fusion.

The results showed that ensemble approaches generally provided more stable performance than individual OCR backbones, particularly under challenging low-resolution conditions. However, improvements obtained from combining predictions were not always consistent across validation and evaluation settings, indicating sensitivity to distribution differences and model agreement behavior. These exploratory experiments were therefore used primarily to guide model selection and understand the strengths and weaknesses of different architectures rather than being included directly in the final methodology.

Insights obtained from this stage contributed to the design of the final system, which moved away from static backbone combinations and adopted a validation-selected track-level ensemble strategy based on agreement, confidence fusion, and character-level evidence aggregation.

3.4 Discussion of the Validation-Test Gap

The validation-test gap is the most important observation in the results. A validation exact-match score above 98% suggests that the model can solve tracks that are similar to the training distribution. The test exact-match score of 79.17% suggests that the hidden test set includes harder examples: lower plate visibility, stronger blur, unusual lighting, more compression, or plate styles and backgrounds less represented in validation. It is also possible that the validation split did not fully match the diversity of Scenario B conditions.

This gap does not mean the system failed. The test character accuracy of 94.53% shows that the OCR backbones still extract useful information from most frames. Instead, it reveals the strictness of exact-match evaluation. For example, for a seven-character plate, one wrong character makes the whole plate incorrect even if six characters are right. Therefore, future improvement should focus on the most ambiguous character positions and on better uncertainty handling rather than only increasing average OCR accuracy.

3.5 Error Sources

- Motion blur: fast-moving cars make character strokes merge or disappear.
- Low spatial resolution: plates may contain too few pixels to distinguish similar letters and digits.
- Lighting and shadows: dark frames and reflections reduce contrast between characters and background.
- Weather and noise: rain, compression artifacts, and sensor noise distort character boundaries.
- Similar characters: 0/O, 1/l, 5/S, 8/B, 2/Z, and 6/G are common sources of near-miss predictions.
- Domain shift: validation tracks may not fully represent hidden test conditions, especially in diverse Scenario B data.

The ensemble reduces some of these errors by combining multiple frames. If one frame is blurred but another frame is clearer, best-frame selection and confidence fusion can recover the correct string. If different models make different mistakes, character-level aggregation can correct the weak positions. However, when all five frames are degraded in the same way, or when every backbone is confused by the same character pair, the ensemble has limited information to recover the correct plate.

3.6 Comparison with Single-Model Reasoning

A single OCR model has two main weaknesses in this project. First, it can become dependent on one visual representation and therefore repeat the same type of error across frames. Second, it may output a high-confidence prediction even when the visual evidence is weak. The multi-backbone ensemble addresses both weaknesses. Different backbones provide different feature biases, and agreement between models becomes a reliability signal. When confidence and agreement disagree, the system can reduce trust in unstable predictions.

The final method also improves over a naive majority vote. Majority voting is strong when the exact same plate string appears many times, but it cannot fully use character-level partial agreement. Our ensemble can choose a candidate that has high support at most character positions even when full-string predictions differ. This is especially useful for low-resolution plates, where only one or two characters are typically uncertain.

3.7 Practical Demonstration

In the intended use case, a surveillance camera captures a moving vehicle over several frames. The detection stage localizes the license plate region, and the recognition system receives five cropped low-resolution plate frames. Each OCR backbone predicts a candidate text and confidence. The ensemble module then compares the outputs. If most models agree on the same plate, that prediction is accepted with high confidence. If predictions differ, the system examines which characters are stable across frames and models, and chooses the most reliable final sequence. The final output is a plate text and confidence score that can be used by a downstream traffic or security system.

Overall, the results support the project hypothesis. Multi-frame evidence and ensemble fusion improve robustness for low-resolution license plate recognition. The remaining challenge is not basic recognition but reliable generalization to the hardest hidden conditions.

The Impact and Future Directions

The outcome of this project has practical impact for real-world surveillance systems where license plates are not captured under ideal conditions. Many existing ALPR systems assume high-resolution, frontal, and well-lit plate images. In contrast, city cameras, campus security cameras, and traffic-monitoring cameras often observe vehicles from a distance and at low resolution. A multi-frame system can make better use of ordinary video footage by combining several weak observations instead of depending on a single image. This can reduce manual review time and improve the reliability of plate-based search, access control, and traffic analysis.

The project also has research impact because it highlights the difference between visual enhancement and recognition-oriented enhancement. Making a plate look sharper is not always the same as improving OCR accuracy. Our results suggest that the most useful direction is to optimize the full recognition pipeline, including temporal evidence, model diversity, and confidence calibration.

If the project is continued, the first future direction should be stronger domain generalization. This can be addressed by adding synthetic degradations such as blur, rain, compression, low light, and random shadows during training. The second direction is temporal modeling. Instead of fusing predictions only at the end, a temporal attention model could learn which frame and which character region should be trusted. The third direction is confidence calibration, so that the confidence score better reflects the true probability of an exact match. The fourth direction is error-aware post-processing for visually similar characters. Finally, a full end-to-end demo could integrate detection, cropping, recognition, and track-level fusion in one interface for real surveillance videos.

References

- Baek, J., Kim, G., Lee, J., Park, S., Han, D., Yun, S., Oh, S. J., & Lee, H. (2019). What is wrong with scene text recognition model comparisons? Dataset and model analysis. ICCV.
- Shi, B., Bai, X., & Yao, C. (2016). An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. CVPR.
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, real-time object detection. CVPR.
- Ledig, C., Theis, L., Huszár, F., Caballero, J., Cunningham, A., Acosta, A., Aitken, A., Tejani, A., Totz, J., Wang, Z., & Shi, W. (2017). Photo-realistic single image super-resolution using a generative adversarial network. CVPR.
- Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., Qiao, Y., & Loy, C. C. (2018). ESRGAN: Enhanced super-resolution generative adversarial networks. ECCV Workshops.
- Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing.
- Huynh-Thu, Q., & Ghanbari, M. (2008). Scope of validity of PSNR in image/video quality assessment. Electronics Letters.
- Zhang, K., Zuo, W., Gu, S., & Zhang, L. (2017). Learning deep CNN denoiser prior for image restoration. CVPR.
- Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. CVPR.
- Chen, X., Wang, X., Zhou, J., Qiao, Y., & Dong, C. (2023). Activating more pixels in image super-resolution transformer. CVPR.
- Du, Y., Li, C., Guo, R., Yin, X., Liu, W., Zhou, J., Bai, X., Yu, Z., Yang, J., & Zhou, J. (2022). SVTR: Scene Text Recognition with a Single Visual Model. IJCAI.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Housley, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR.

- Gu, A., & Dao, T. (2023). Mamba: Linear-Time Sequence Modeling with Selective State Spaces. arXiv preprint arXiv:2312.00752.
- Zhang, Y., Peng, D., Kong, Y., Liu, Y., Wang, L., & Jin, L. (2020). Contextualized Scene Text Recognition with Transformer. IJCAI.
- Breiman, L. (2001). Random Forests. Machine Learning.
- Geurts, P., Ernst, D., & Wehenkel, L. (2006). Extremely Randomized Trees. Machine Learning.
- Dietterich, T. G. (2000). Ensemble methods in machine learning. Multiple Classifier Systems.